

SESSION 2

RAD-541 Magnetic Resonance Imaging II

Gradient Fields, Imaging Axes and Slice Selection

How MRI makes resonance frequency depend on position and selects a specific slice

Course	Magnetic Resonance Imaging II
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Suggested study time	240–300 minutes

Learning Objectives

By the end of this session, you should be able to:

1. explain why a homogeneous main field, B_0 , cannot by itself localize an MR signal;
2. define a magnetic-field gradient and interpret gradient strength in mT/m;
3. distinguish the physical x , y and z gradient axes from the functional slice, phase and frequency/readout directions;
4. explain how a slice-select gradient and a frequency-selective RF pulse work together to excite one slice;
5. predict how RF centre frequency, RF transmit bandwidth and slice-select gradient strength affect slice position and thickness;
6. explain how axial, sagittal, coronal and oblique planes are produced; and
7. distinguish 2D slice selection from through-plane or partition encoding in 3D imaging.

The central idea

A homogeneous B_0 gives hydrogen a stable reference resonance frequency, but it provides almost no information about *where* a signal came from. MRI solves this by switching on small, controlled **gradient fields**. A gradient makes magnetic-field strength vary predictably with position; because Larmor frequency follows field strength, resonance frequency also becomes position-dependent. MRI can then use frequency to select a slice and, later, to help encode the remaining spatial directions.

1 Session Orientation

Session 1 established the Larmor relationship: magnetic-field strength determines precessional frequency. Session 2 applies that relationship to localization. The logic is:

$$\begin{aligned} B_0 &\longrightarrow \text{stable reference frequency} \longrightarrow \text{controlled gradient field} \\ &\longrightarrow \text{position-dependent frequency} \longrightarrow \text{RF selects a slice.} \end{aligned}$$

The most important distinction in this session is between **physical gradient axes** and **functional imaging directions**. The scanner owns three physical gradient systems, but the sequence assigns those systems different jobs depending on the prescribed imaging plane.

2 Why B_0 alone cannot localize signal

If every hydrogen proton in the imaging volume experienced exactly the same magnetic field, identical protons would precess at essentially the same Larmor frequency. That uniformity is excellent for establishing resonance, but poor for localization: a signal received at that frequency would not tell the scanner whether it came from the head, chest or abdomen.

Spatial encoding therefore begins by adding a small field whose magnitude changes with position. This additional field does not replace B_0 ; it is superimposed on it. At one position it may slightly increase the total field, while at another it may slightly decrease it.

A useful mental model

Think of B_0 as the **stable reference field**. Gradients are temporary, controlled distortions of that reference. The scanner knows the gradient amplitude and direction, so the resulting frequency changes are useful rather than accidental.

3 What is a magnetic-field gradient?

A **magnetic-field gradient** is a controlled change in magnetic-field strength with distance. Over the imaging region, the change is designed to be approximately linear. For a gradient applied along the z direction,

$$B(z) = B_0 + G_z z,$$

where G_z is the gradient strength. Gradient amplitude is commonly expressed in **millitesla per metre (mT/m)**: it tells us how much the magnetic field changes for each metre of position.

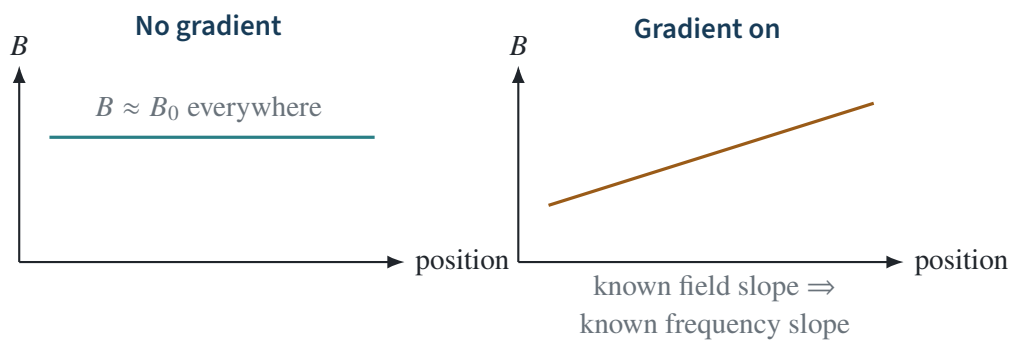
Because Larmor frequency is proportional to field strength,

$$f(z) = \bar{\gamma} B(z) = \bar{\gamma} (B_0 + G_z z),$$

where $\bar{\gamma} = \gamma/2\pi \approx 42.58 \text{ MHz/T}$ for hydrogen.

The physical consequence is the key point:

different position $\rightarrow$ different magnetic field $\rightarrow$ different Larmor frequency.



Why this matters to the radiographer

Every prescribed slice, plane change, phase/frequency swap and many spatial-resolution choices ultimately rely on gradient fields. The console may hide the waveform details, but gradients are the hardware that creates image geometry.

4 The three physical gradient axes

Clinical MRI systems contain three orthogonal gradient-coil sets. These are conventionally labelled G_x , G_y and G_z . In a standard cylindrical scanner, z is along the bore; x and y form the two perpendicular transverse directions. Patient-relative labels such as left–right and anterior–posterior depend on positioning and coordinate convention, so the safest concept is that x , y and z are the scanner’s **physical gradient coordinates**.

During an acquisition, however, gradients are assigned **functional roles**:

Slice-select direction

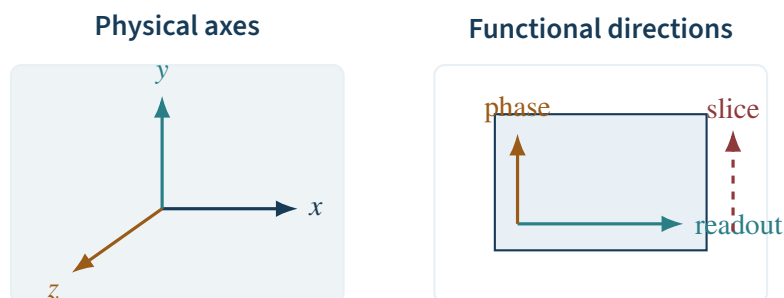
The direction perpendicular to the excited slice or slab.

Frequency/readout direction

One in-plane direction, encoded mainly while signal is sampled.

Phase direction

The other in-plane direction, encoded using accumulated phase differences.



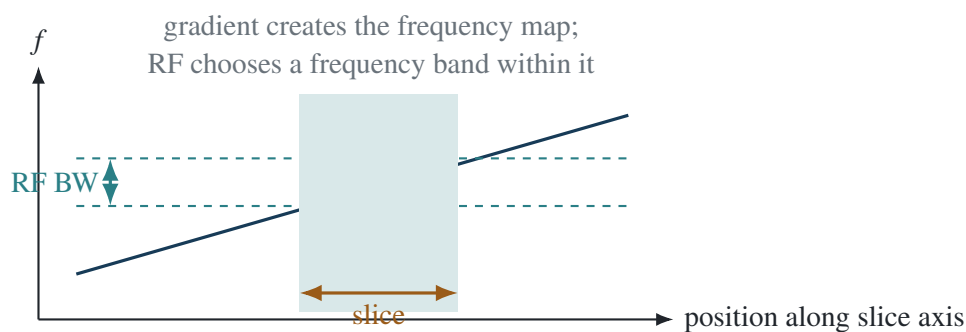
Physical axes are not permanently assigned to one job

Do not memorize “ G_z is slice, G_x is frequency and G_y is phase” as a universal rule. $x/y/z$ describe physical coil coordinates. Slice/phase/frequency describe jobs assigned for a particular sequence and plane. The scanner can combine physical gradient outputs to create the required resultant direction.

5 How MRI selects a slice

If an RF pulse were transmitted while B_0 remained perfectly homogeneous, spins over a very large region could share the same resonance frequency and be excited together. To select a thin region, MRI switches on a **slice-select gradient** during RF excitation.

The gradient creates a predictable position–frequency map along the chosen slice direction. At the same time, the RF pulse contains a limited range of frequencies—its **transmit bandwidth**. Spins whose Larmor frequencies fall within that RF band are efficiently excited; spins outside it are not part of the intended slice excitation.



5.1 Slice selection step by step

1. The patient is already in B_0 and hydrogen spins have their local Larmor frequencies.
2. The slice-select gradient is switched on along the desired slice direction.
3. Magnetic-field strength now changes with position along that direction.
4. Larmor frequency therefore changes with position.
5. An RF pulse is transmitted with a selected centre frequency and transmit bandwidth.
6. Spins whose frequencies lie within that RF band are excited strongly.
7. That frequency band corresponds to a spatial band: the selected slice.

The gradient does not excite the slice

The slice-select gradient does not itself tip the net magnetization into the transverse plane. The **RF pulse provides excitation**. The gradient's job is to make resonance frequency depend on position so that the RF frequencies correspond to a particular spatial region.

6 Slice position: the role of RF centre frequency

While the slice-select gradient is on, each location along the slice axis corresponds to a particular resonance frequency. The **centre frequency** of the RF pulse therefore determines the centre position of the selected slice.

If the RF centre frequency is shifted while gradient strength and bandwidth stay the same, the slice moves along the gradient direction without necessarily changing thickness. Reversing gradient polarity reverses the direction in which frequency changes with position, so the scanner coordinates RF frequency and gradient polarity automatically when positioning a slice stack.

What this means on the console

Moving a slice or slice stack on the localizer ultimately changes the relationship between gradient field and RF centre frequency. The radiographer prescribes anatomy; the scanner calculates the required gradient/RF combination.

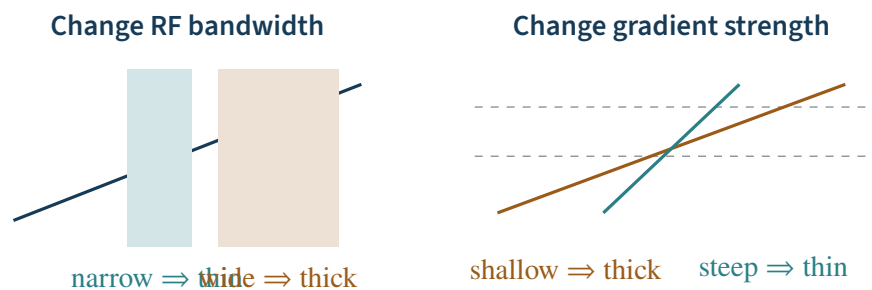
7 Slice thickness: bandwidth and gradient strength

Slice thickness depends mainly on the **RF transmit bandwidth**, Δf_{RF} , and the magnitude of the slice-select gradient, $|G_{SS}|$. For hydrogen,

$$\Delta z = \frac{\Delta f_{RF}}{\gamma |G_{SS}|}.$$

This relationship gives two high-yield rules:

- wider RF transmit bandwidth $\rightarrow$ thicker slice, if gradient strength is unchanged;
- stronger/steeper slice-select gradient $\rightarrow$ thinner slice, if RF bandwidth is unchanged.



Slice-thickness calculation

Suppose the RF transmit bandwidth is 2 kHz and the slice-select gradient is 10 mT/m = 0.010 T/m. Using $\gamma = 42.58 \times 10^6$ Hz/T:

$$\Delta z = \frac{2000}{(42.58 \times 10^6)(0.010)} \approx 0.0047 \text{ m} = 4.7 \text{ mm}.$$

If the gradient strength is doubled while the RF bandwidth remains unchanged, the slice thickness is approximately halved.

Transmit bandwidth is not receiver bandwidth

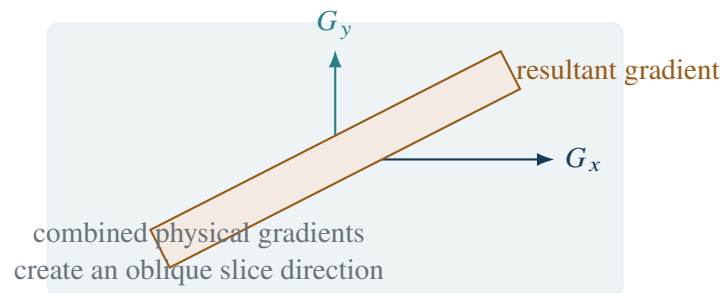
The bandwidth in the slice-thickness equation is the **RF transmit bandwidth**: the range of frequencies present in the excitation pulse. **Receiver bandwidth** describes the frequency range sampled during signal readout and is discussed with frequency encoding. The two parameters are related to frequency, but they perform different jobs.

8 Axial, sagittal, coronal and oblique planes

The slice-select direction is perpendicular to the image plane. In simple orthogonal examples:

Desired plane	Approximate slice-select direction	In-plane directions
Axial	head–foot	left–right and anterior–posterior
Sagittal	left–right	anterior–posterior and head–foot
Coronal	anterior–posterior	left–right and head–foot

Modern MRI is not limited to these three planes. By energizing two or three physical gradient systems simultaneously, the scanner creates a **resultant gradient** at an oblique angle. The resultant slice-select gradient is perpendicular to the prescribed oblique plane.



Radiographer's plane prescription

When you angle a slice stack on localizer images, the gradient coils do not physically rotate. The system calculates the required combination of G_x , G_y and G_z so that the resultant gradient points perpendicular to the plane you prescribed. This is fundamental to cardiac, joint, spine and neurovascular imaging.

9 2D slice selection versus 3D partition encoding

The phrase “slice encoding” can be confusing because 2D and 3D acquisitions treat the through-plane direction differently.

In a conventional **2D acquisition**, RF excitation plus a slice-select gradient excites one relatively thin slice (or a sequence of individual slices). The two directions inside that slice are then encoded by phase and frequency.

In a typical **3D acquisition**, the scanner excites a thicker slab. Position through the slab is then resolved using an additional phase-encoding dimension, often called **partition encoding** or **through-plane phase encoding**. The reconstructed thin sections are therefore produced from the encoded 3D volume rather than being individually selected in the same way as 2D slices.

Slice selection is not partition encoding

Slice selection uses RF plus a gradient to determine the excited region. Partition encoding is an additional spatial-encoding step used to distinguish locations within a 3D slab. They are related concepts, but they are not the same operation.

10 Gradient performance: amplitude and slew rate

Two hardware terms are worth knowing:

Gradient amplitude

How steeply the field changes with distance, usually expressed in mT/m.

Slew rate

How quickly gradient amplitude can change with time, usually expressed in T/m/s.

Higher gradient performance supports demanding spatial encoding and rapid sequences, but gradient switching is constrained by hardware limits, acoustic noise and patient-safety considerations such as peripheral nerve stimulation.

Practical meaning

Radiographers rarely enter gradient amplitude or slew rate directly. The scanner converts choices such as slice thickness, FOV, bandwidth and sequence timing into gradient waveforms. Understanding the physics helps you predict the direction of parameter effects even when the console hides the actual waveform values.

11 Putting the spatial framework together

The spatial-localization chain is now:

1. B_0 provides a predictable baseline Larmor frequency.
2. Gradients make field strength and frequency depend on position.
3. RF plus the slice-select gradient chooses a slice or slab.
4. The two remaining in-plane directions must still be encoded by frequency and phase.

That final step is the focus of Session 3.

Key Takeaways

- Gradients create controlled field variation so that Larmor frequency can carry spatial information.
- G_x , G_y and G_z are physical gradient axes; slice, phase and frequency are functional directions assigned for a particular acquisition.
- Slice selection requires a slice-select gradient and a frequency-selective RF pulse acting together.
- RF centre frequency controls slice position along the gradient direction.
- Wider transmit bandwidth produces a thicker slice when gradient strength is fixed; a stronger slice-select gradient produces a thinner slice when RF bandwidth is fixed.
- Oblique imaging is produced by combining physical gradient outputs into a resultant gradient direction.
- 2D slice selection and 3D partition encoding are related but distinct processes.

Key Terms

Gradient field

Controlled variation of magnetic-field strength with position.

Gradient amplitude

Field change per unit distance, usually mT/m.

Physical gradient axes

Scanner-coordinate gradient directions x , y and z .

Slice-select gradient

Gradient used during RF excitation to make resonance frequency position-dependent along the slice direction.

Transmit bandwidth

Frequency range present in the excitation RF pulse.

Slice thickness

Spatial width of the excited region along the slice-select direction.

RF centre frequency

Central excitation frequency that determines the centre position of the selected slice for a given gradient.

Frequency/readout direction

In-plane direction encoded by frequency during signal acquisition.

Phase direction

In-plane direction encoded using phase differences.

Oblique imaging

Imaging plane created by combining physical gradients into an angled resultant gradient.

Partition encoding

Through-plane phase encoding used to resolve positions within a 3D slab.

Slew rate

Rate at which gradient amplitude changes with time.

Self-Assessment Questions

1. Why is a homogeneous B_0 alone insufficient for spatial localization?
2. Define gradient strength and state its common unit.
3. Distinguish physical $x/y/z$ gradient axes from slice, phase and frequency directions.
4. Explain step by step how a slice-select gradient and RF pulse select one slice.
5. What happens to slice position when RF centre frequency is changed while gradient strength remains fixed?
6. What happens to slice thickness when RF transmit bandwidth increases while gradient strength is unchanged?
7. What happens when the slice-select gradient becomes steeper while RF bandwidth is unchanged?
8. How can three physical gradient systems produce an oblique imaging plane?
9. Explain the difference between 2D slice selection and 3D partition encoding.
10. A sequence uses a 2 kHz RF bandwidth and a 10 mT/m slice-select gradient. Using $\bar{\gamma} = 42.58 \text{ MHz/T}$, estimate the slice thickness.

Series navigation

Next session: Frequency Encoding and Phase Encoding. With one slice selected, MRI still has to determine where signal originated within that slice. Session 3 explains how the readout gradient assigns frequency along one in-plane direction and how the phase-encoding gradient assigns phase along the other.

